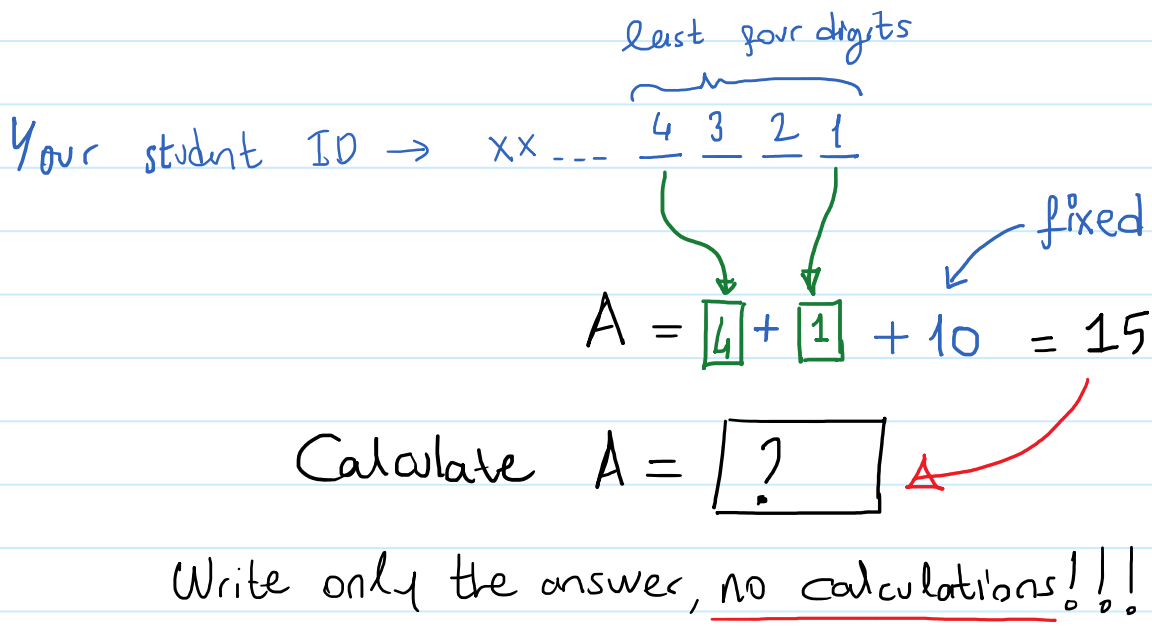
Q1

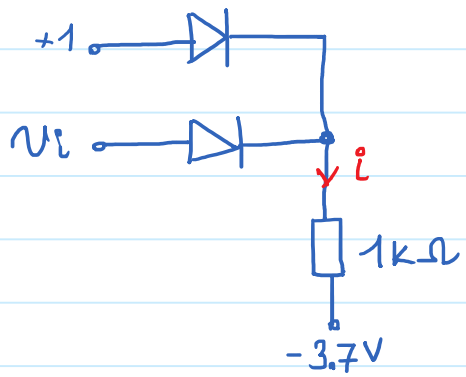


Write only the answer, no calculations!!!

Q2

← the value you calculated in Q1

$V_i = A \cos \omega t$



Calculate

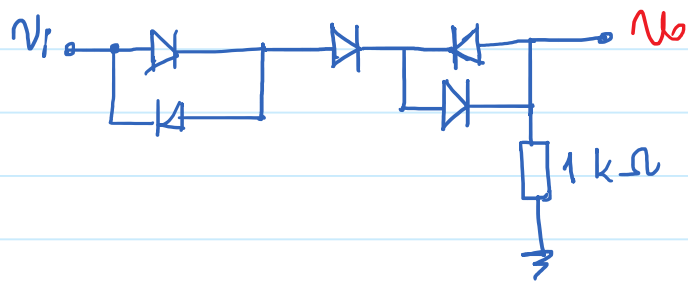
$\max(i) - \min(i) = ?$

Use constant voltage drop model.
Don't forget to write units.

Q3

← the value you calculated in Q1

$V_i = A \cos \omega t$



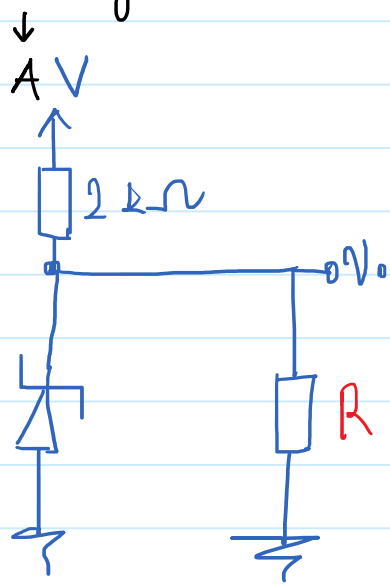
Calculate $\max(V_o) - \min(V_o)$

Use constant voltage drop model. Don't forget to write units.

Q4

A zener diode is rated for 5 V at 3mA current with an incremental resistance of 100Ω.

the value you calculated in Q1



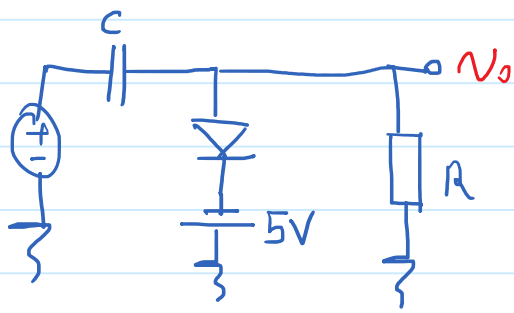
What is the max value of R such that $V_o \geq 4.8$ V?

Use constant voltage drop model. Don't forget to write units.

Q5

← the value you calculated in Q1

$V_i = A \cos 2\pi f t$



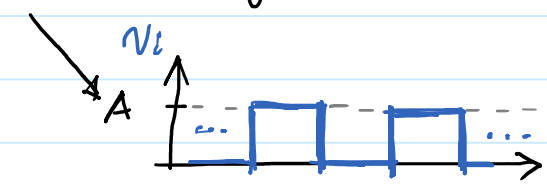
Assume $RC \gg \frac{1}{f}$

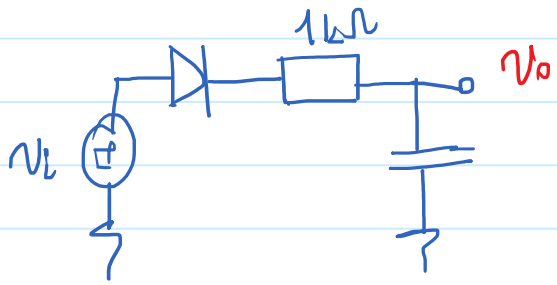
Calculate $\left| \max(V_o) + \min(V_o) \right| = ?$ ← absolute value

Use constant voltage drop model. Don't forget to write units.

Q6

the value you calculated in Q1





Find $\max(v_o) = ?$

Use constant voltage drop model. Don't forget to write units.